

AHMED ADHAM | RESUME

- » **University:** New Cairo Technological University
- » **Skills:** Python, Machine Learning, Deep Learning, NLP, Data Structures & Algorithms, FastAPI, React, C++/C#, Java
- » **Interests:** Artificial Intelligence, Machine Learning, Deep Learning, Quantum Computing, Quantum Machine Learning

»»» Summary

Computer Science graduate specializing in Software Development, with a strong interest in Machine Learning, Deep Learning, Artificial Intelligence, and Quantum Computing. Experienced in developing AI-driven solutions, intelligent systems, and data-focused applications through academic projects, research, and hackathons. Passionate about exploring advanced computational technologies and applying them to solve complex real-world problems.

»»» Education

2022 - 2026 **Bachelor's Degree in Information Technology** New Cairo Technological University

- » Major in Software Development.

»»» Experience

2026 **Software Developer, Skillable Initiative (Graduation Project)** New Cairo Technological University

- » Co-developed *Skillable Initiative*, a full-stack accessibility-first recruitment platform connecting job seekers with disabilities and recruiters through intelligent job matching.
- » Designed an explainable weighted matching system incorporating accessibility requirements, skills, experience, and contextual factors, supported by skill graphs and rule-based symbolic AI.
- » Implemented accessibility features including high-contrast modes, font scaling, speech-on-focus, keyboard navigation, and bilingual Arabic/English support with RTL layouts.
- » Built a modular architecture using React and FastAPI with dedicated matching and authentication services, secured using JWT authentication and Argon2 password hashing.

Apr 2026 **Machine Learning Engineer, DeepX Hackathon** Nile University, Egypt

- » Developed an Arabic Aspect-Based Sentiment Analysis (ABSA) system to analyze customer reviews across multiple aspects including food, service, and pricing.
- » Implemented multi-aspect sentiment classification pipelines and optimized preprocessing workflows for Arabic NLP tasks under a 12-hour hackathon deadline.
- » Introduced a *not_mentioned* classification strategy to prevent forced sentiment predictions for absent aspects, significantly improving model accuracy and evaluation performance.
- » Worked on real-world Arabic customer review datasets using machine learning and NLP techniques for fine-grained sentiment understanding.

Feb 2026 **Co-Founder & Software Developer, Seshat Systems** Benha City, Egypt

- » Co-developed *Seshat*, a FHIR-compliant digital health platform transforming fragmented clinic systems into a connected ecosystem.
- » Implemented an AI-powered drug interaction engine, real-time lab results viewer, and *Wadjet* — an ICD-10 coding assistant to improve diagnostic accuracy and automate insurance workflows.
- » Focused on improving healthcare outcomes by reducing medication errors, accelerating diagnostics, and enabling continuity of care.

Dec 2025	Software Developer, GEM Hackathon (Hack054)	Grand Egyptian Museum, Egypt
	» Developed a mobile application to support visitor experience analysis at the Grand Egyptian Museum using Bluetooth beacon-based indoor tracking. » Implemented a system to track visitor movement patterns, enabling data-driven insights into user behavior and museum engagement. » Designed an artifact bookmarking and recommendation feature to personalize visitor experiences based on user preferences. » Built a real-time backend system (<i>GEM Database Monitor</i>) to visualize live user interactions and continuously update analytics dashboards.	
Oct 2025	Game Developer, Stellar Lands (NASA Space Apps Cairo 2025)	zewail city of science and technology
	» Developed <i>Stellar Lands</i>, a farming simulator game built with the Godot engine for the NASA Space Apps Cairo 2025 challenge "<i>Farmer Navigator</i>". » Implemented GIS-based terrain mapping and problem visualization to simulate environmental and agricultural conditions. » Designed mathematical growth models and simulation logic to realistically represent real-world planting dynamics. » Collaborated within a multidisciplinary team to integrate gameplay mechanics, balance systems, and deliver a polished, playable prototype by the submission deadline. » Created pixel-art visuals and interactive features to enhance scientific engagement and educational value.	
Sep 2025	Researcher, Quantum LSTM Air Quality Prediction	Nile University
	» Developed a Quantum LSTM (QLSTM) model to predict PM2.5 concentrations, validating results on real IBM quantum hardware with performance comparable to classical models. » Designed a hybrid GNN-LSTM spatio-temporal architecture to compress high-dimensional environmental data into efficient temporal representations.	
Jul 2025	AI Developer, ExamAi (EduHackAI-1, 3rd Place)	Sahaba Club
	» Developed a bilingual smart exam monitoring system that enhances academic integrity through AI-based real-time proctoring. » Implemented facial and eye tracking using OpenCV, voice detection with SoundDevice, and activity logging with Pynput. » Integrated periodic snapshots, full-session recording, and real-time alerts with a multi-lingual (Arabic/English) GUI. » Built with Python, Tkinter, OpenCV, SoundDevice, Pandas, and Threading; tested successfully in a live exam simulation.	
Dec 2024	Robotics Project	New Cairo Technological University
	» Built an Arduino-based line-following robot. » Handled both coding and physical construction, including wiring and sensor connections.	
Nov 2024	Project Management Trainee	American Chamber of Commerce in Egypt
	» Completed a professional project management training focused on leadership, workflow optimization, and project lifecycle execution. » Gained hands-on experience in planning, coordination, and team management aligned with PMP best practices.	
Nov 2023 - Jul 2024	Frontend Developer, Instant Booking (2nd-Year College Project)	New Cairo Technological University

- » Developed a responsive and user-friendly interface for a course booking platform.
- » Implemented animations and styling using CSS and Bootstrap.

»»» Certifications

May 2026	Introduction to Deep Learning & Neural Networks with Keras	IBM
	» Completed foundational training in deep learning, neural networks, backpropagation, and model optimization using Keras and TensorFlow. » Developed a final project using a pre-trained VGG16 model for aircraft damage classification and a Transformer-based model for automated image caption generation.	
May 2026	AWS Academy - Machine Learning Foundations	Amazon Web Services (AWS)
	» Completed foundational training in machine learning concepts, model development workflows, and practical AI applications using AWS technologies. » Explored supervised learning, computer vision, natural language processing, and responsible AI fundamentals.	
2025	Machine Learning & Python for AI(36 hrs)	Black Horse Academy
	» Completed an applied course focused on both supervised and unsupervised machine learning techniques. » Worked on hands-on projects using Python, NumPy, Pandas, and TensorFlow for model training and evaluation.	
2025	MEAN Stack Development (60 hrs)	Information Technology Institute (ITI)
	» Completed an intensive full-stack web development course covering MongoDB, Express.js, Angular, and Node.js. » Gained practical experience building dynamic web applications and RESTful APIs.	
2025	Quantum Summer School (2 Weeks)	Nile University
	» Explored quantum computing concepts and their applications in machine learning. » Participated in a Quantum Machine Learning challenge focused on air quality prediction.	
2024	Networking Principles	Radio Television Institute (RTI)
	» Studied computer networks, routing, switching, and troubleshooting techniques.	
2024	Project Management Professional (PMP)	American Chamber of Commerce
	» Trained on leadership, project planning, workflow optimization, and professional project execution.	

»»» Activities & Achievements

Apr 2026	DeepX Hackathon — 11th Place	Nile University, Egypt
	» Achieved 11th place out of 151 competing teams in a 12-hour AI hackathon focused on Arabic Aspect-Based Sentiment Analysis (ABSA). » Built a fine-grained sentiment analysis system capable of identifying sentiments for specific review aspects rather than overall review polarity.	
Feb 2026	Benha Healthcare Hackathon — 2nd Place	Benha City, Egypt
	» Co-founded <i>Seshat</i>, a healthcare platform connecting clinics, doctors, and patients into a unified system. » Delivered a 20-minute pitch in the medical track among competing healthcare teams.	
Dec 2025	GEM Hackathon (3rd Edition)	Grand Egyptian Museum, Egypt

	» Participated as part of team <i>Hack054</i>, developing a smart museum application for visitor tracking and personalized recommendations.	
2025	NASA Space Apps Cairo	Cairo, Egypt
	» Developed a playable prototype of <i>Stellar Lands</i> during the hackathon.	
2025	EduHackAI-1 — 3rd Place	Dubai, UAE
	» Achieved 3rd place for the project <i>ExamAi</i>.	
2025	Quantum Summer School (2 Weeks)	Nile University
	» Explored quantum computing concepts and their applications in machine learning. » Participated in a Quantum Machine Learning challenge focused on air quality prediction.	
2024	Quantum AI Hackathon	Nile University
	» Worked on team-based quantum computing projects, developing innovative AI solutions.	
2024	Google Developer Groups Cairo	Cairo, Egypt
	» Attended workshops and hackathons, expanding practical knowledge in software development and AI.	